

status, risk aversion, and all the other important considerations of a client—all the factors in an agent's life cycle.

4.4. ANNUITIES

Annuities should be a retiree's best friend. They provide steady payments, there is no risk of a retiree outliving his savings, and some versions index payments to inflation.

The beauty of annuities is that you receive other people's money. The best part is that they don't miss it because they are dead, so everyone is better off. This intuition was first shown in a seminal paper by Menahem Yaari (1965), an economist at Hebrew University. Yaari showed that an annuity is like an insurance policy with a negative price (in fact, annuities are reverse death insurance): those who die early subsidize the consumption of the lucky Methuselahs who die late. But everyone benefits because at the start you don't know if you're going to die early or late, and you can't consume once you die. Yaari demonstrated there were gigantic benefits of investing in annuities.

The original setting by Yaari was restrictive: he assumed the annuities were fairly priced, there were no fees or frictions, individuals didn't want to save anything to pass onto their heirs, and there were no uninsurable risks. Taking these into account reduces the attractiveness of annuities, but there are still large benefits. Yogo (2011) includes health shocks, bequest motives, and other considerations and shows that there are gains of more than 15% of household financial wealth if households were to purchase additional annuities (over those already embedded in Social Security).⁴⁷

You never outlive your money with an annuity. The risk of an individual outliving her savings has been steadily increasing because lifespans have been getting longer. The Society of Actuaries reports that life spans have improved by around two years each decade for males, from 66.6 years in 1960 to 75.7 years in 2010, and around 1.5 years per decade for females, from 73.1 years in 1960 to 80.8 years in 2010.⁴⁸ "There is a general misunderstanding of what 'average life expectancy' means, and when people are told that they will live to an age such as eighty or eighty-five, they don't realize that this means there is a 50% chance that they could live past that age," says an actuary. Many people don't even get the average right, with 41% underestimating their life expectancies by at least five years. The longevity risk is substantial: by age sixty-five, U.S. males in average health have a

⁴⁷Davidoff, Brown, and Diamond (2005) also showed large benefits of annuitization even with health-care risk or when available annuity products could not exactly meet agents' desired consumption.

⁴⁸Numbers and quotation are from McKeown, K., and E. Michalak, "Retirees Underestimate Life Expectancy, Risk Underfunding Retirement," Society of Actuaries press release, July 30, 2012. The full report is Society of Actuaries, 2011 Risks and Process of Retirement Survey.